

SUMMARY

Software engineer with proven ability to ship production systems using Python ? TypeScript ? and JavaScript. Strong object-oriented programming fundamentals with hands-on experience in distributed systems architecture. Effectively leverage AI coding tools (Claude Code ? Cursor ? GitHub Copilot) to accelerate development. Passionate about writing well-tested ? maintainable code.

CORE COMPETENCIES

OOP

Python

TypeScript

TDD

System Design

WORK EXPERIENCE

OpenPRA Org

Aug 2025 ? Present

Software Engineer

Owned end-to-end delivery of a probabilistic fault-tree generator for nuclear safety analysis ? shipped from concept to production in 8 weeks using Python ? TypeScript ? Node.js ? and Docker with full CI/CD automation.

Authored 25+ Jest unit tests and CI checks guaranteeing numerical stability and reproducible outputs ? maintaining full coverage on safety-critical probability calculations.

Architected type-safe probability classes across packages ? enabling rapid testing and iteration loops.

Redesigned XML processing pipeline with lazy-parse techniques ? cutting debugging cycles by 40% and accelerating risk-assessment turnaround.

NC State University

Jul 2024 ? Mar 2025

Backend Engineer ? Research Assistant

Designed and deployed a production RAG stack using LangChain ? FastAPI ? FAISS ? HuggingFace transformers ? and Redis ? achieving 85% extraction accuracy and sub-200ms latency under concurrent load.

Engineered retrieval pipeline with semantic chunking ? confidence thresholds, and exponential-backoff retry logic ? eliminating 60% of manual work and saving the department 30 hours/week.

Integrated distributed tracing and structured logging into the FastAPI service; used telemetry to identify bottlenecks and drive a 30% throughput improvement.

Resilient Tech

Jan 2023 ? May 2023

Software Engineer

Migrated critical ERP systems to RESTful Python APIs across 5 enterprise rollouts ? cutting production downtime by 30% and sustaining 99% uptime.

Resolved 100+ production support tickets by tracing root causes through front-end and back-end workflows ? lifting CSAT by 12%.

Built and maintained client-facing React interfaces alongside Python backend APIs ? delivering full-stack features using Git-based version control and Agile workflows.

PROJECTS

Restaurant Voice Hub | Python ? FastAPI ? PostgreSQL ? ElevenLabs

Architected agentic orchestration with 5 callable tools for autonomous voice ordering.

Engineered real-time PostgreSQL sync via CSV pipeline ? cutting menu update to seconds.

Deployed multi-tenant architecture across Vercel/Render ? ready for scaling.

EDUCATION

M.S. Computer Science, North Carolina State University

May 2025

GPA: 3.63/4

B.Tech Computer Engineering, Pandit Deendayal Energy University

May 2023

GPA: 9.62/10

TECHNICAL SKILLS

Languages: Python ? TypeScript ? JavaScript ? Java ? SQL

Frameworks: FastAPI ? Flask ? Django ? Node.js ? React

Data & Cloud: PostgreSQL ? Redis ? AWS ? GCP ? Docker

AI Tools: Claude Code ? Cursor ? GitHub Copilot